Non-Profit Support Triage Agent

Course Name : Datagami Skill Based Course

Institution Name: *Medicaps University*

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Project Number: AAI-35

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Academic Year: 2025-26

Problem Statement & Objectives

1. Problem Statement

Non-profit organizations receive a large volume of incoming emails such as donation inquiries, complaints, emergency assistance requests, and insurance claims. These communications are typically unstructured and vary in intent and urgency, making them difficult to process efficiently.

Manual handling of such emails is time-consuming, error-prone, and often leads to delays in response. In critical situations, delayed responses can impact timely assistance and reduce stakeholder satisfaction.

To address these challenges, there is a need for an intelligent automated system that can fetch emails directly from communication channels, analyze their content, classify intent, determine urgency, extract key information, and generate appropriate responses in real time.

The system is designed to integrate with Gmail to automatically retrieve incoming emails, process them using AI models, and send context-aware responses back to the sender, enabling a fully automated email triaging workflow.

2. Project Objectives

Primary Objective

The primary objective of this project is to design and develop an AI-powered system that automates the triaging and handling of incoming communications for non-profit organizations.

The key objectives include:

1. To develop an intelligent system capable of understanding unstructured text messages
2. To accurately classify the intent of messages (e.g., donation, complaint, emergency request)
3. To detect the urgency level of each message to prioritize responses
4. To extract important entities such as names, dates, locations, and monetary values
5. To generate context-aware and professional responses using generative AI
6. To reduce manual workload and improve operational efficiency
7. To ensure faster response time and better user experience
8. To integrate email services (IMAP & SMTP) for automatic email fetching and response delivery

3. Scope of the Project

Included:

- Development of a web-based application for monitoring and processing emails
- Automatic fetching of emails from Gmail inbox using IMAP
- Real-time processing of incoming emails
- AI/LLM-based natural language processing for text analysis
- Automated intent classification, urgency detection, and entity extraction
- Automated response generation using LLM
- Automatic reply sending system using SMTP
- Rule-based decision engine for improving reliability
- Simple and user-friendly interface for triggering and monitoring processing

Excluded:

- Persistent data storage or database integration
- Large-scale deployment and cloud infrastructure setup
- Advanced analytics dashboards or reporting tools
- Integration with multiple communication platforms beyond email

Proposed Solution

1. Key Features

- Automated intent classification of incoming emails
- Urgency detection to prioritize critical requests
- Entity extraction for structured information generation
- AI-based response generation using Large Language Models

• Rule-based decision engine for refining outputs

• Real-time processing pipeline

• Simple and user-friendly web interface

2.Overall Architecture / Workflow

The system follows a modular and pipeline-based architecture where each component performs a specific task.

Workflow Steps:

1. The system connects to Gmail using IMAP
2. The latest email is automatically fetched from the inbox
3. The email content is passed to the Triage Agent (LLM Service)
4. The system performs:
   - Intent classification
   - Urgency detection
   - Entity extraction
5. The Decision Engine refines and validates outputs using rule-based logic
6. The processed data is passed to the Response Generator
7. A professional and context-aware response is generated
8. The system sends the generated response back to the sender using SMTP
9. The final results are displayed on the web interface

This workflow enables a fully automated, real-time email processing and response system.

3. Tools & Technologies Used

- Python – Core programming language for backend processing
- Streamlit – Used to build an interactive and user-friendly web interface
- Groq API – Provides access to high-performance LLM inference
- LLaMA Models – Used for natural language understanding and response generation
- Natural Language Processing (NLP) – For text analysis, classification, and extraction
- Rule-Based Logic – For refining AI outputs and ensuring reliability
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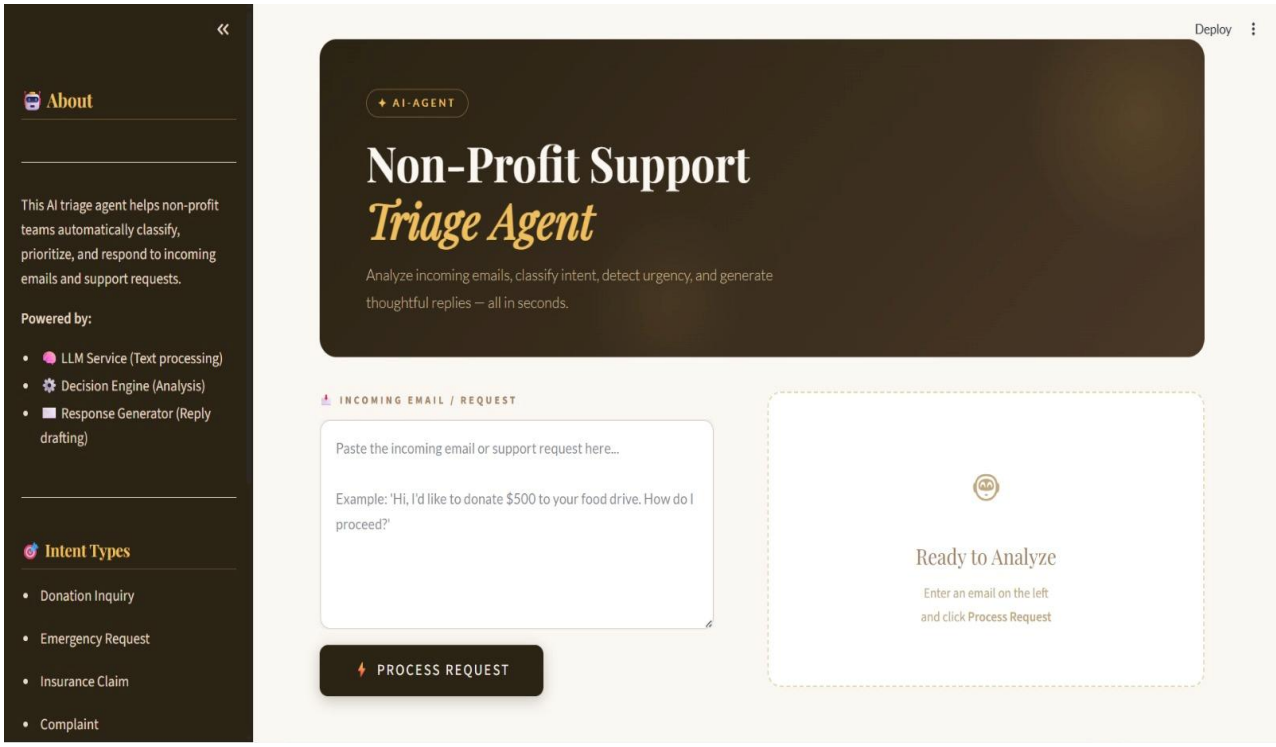
- IMAPClient – For fetching emails from Gmail inbox
- SMTP – For sending automated email responses

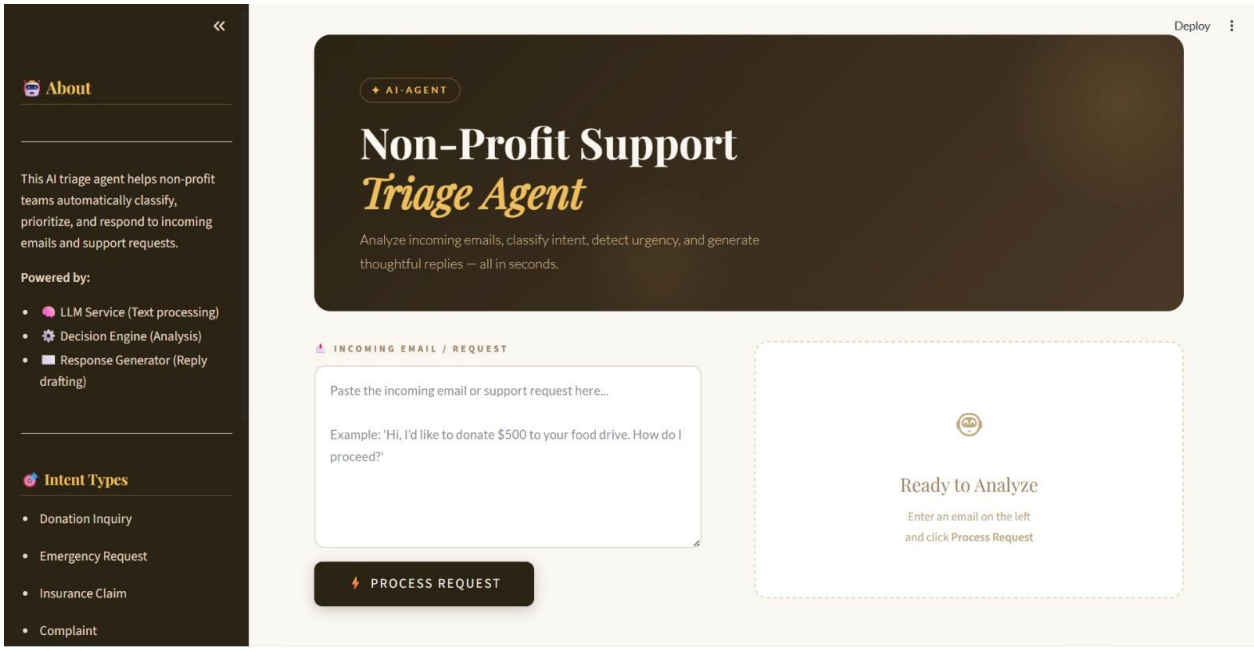
Results & Output

Screenshots / Outputs

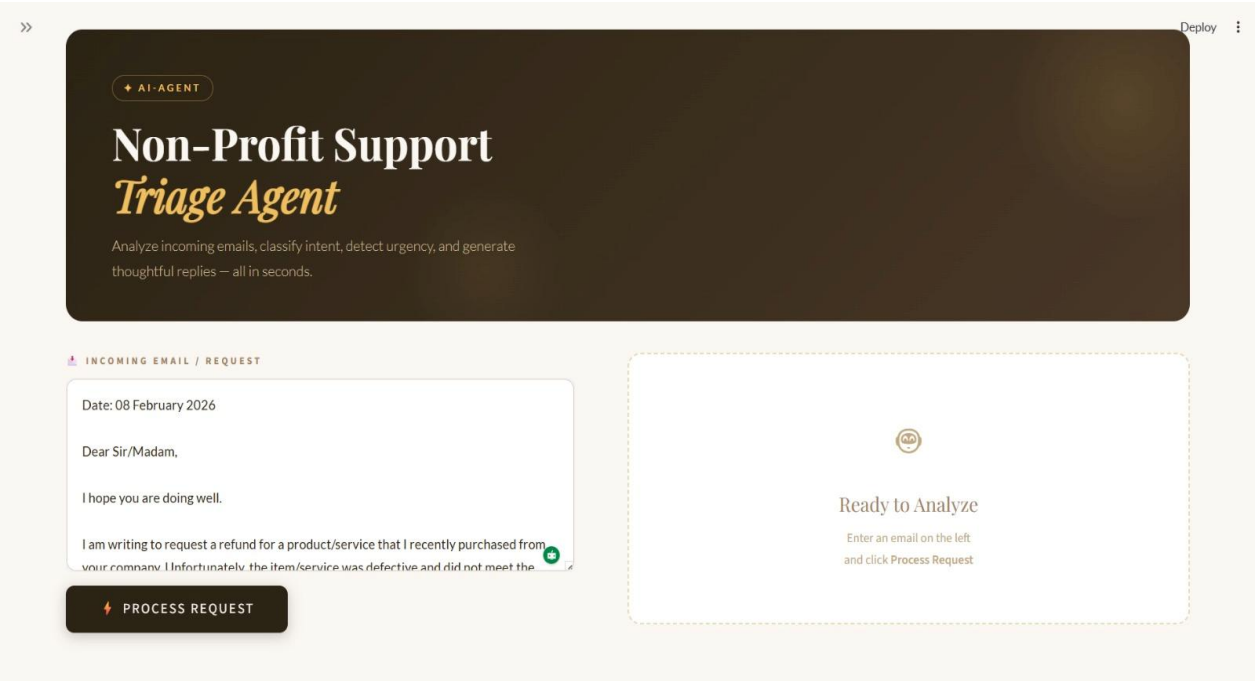
1. UI Interface



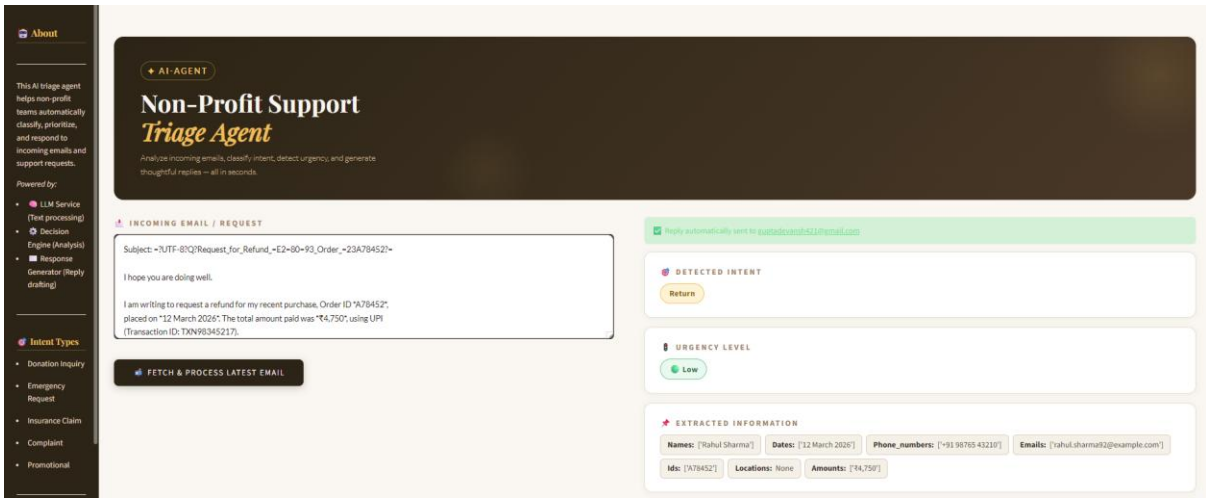
2. Message Interface

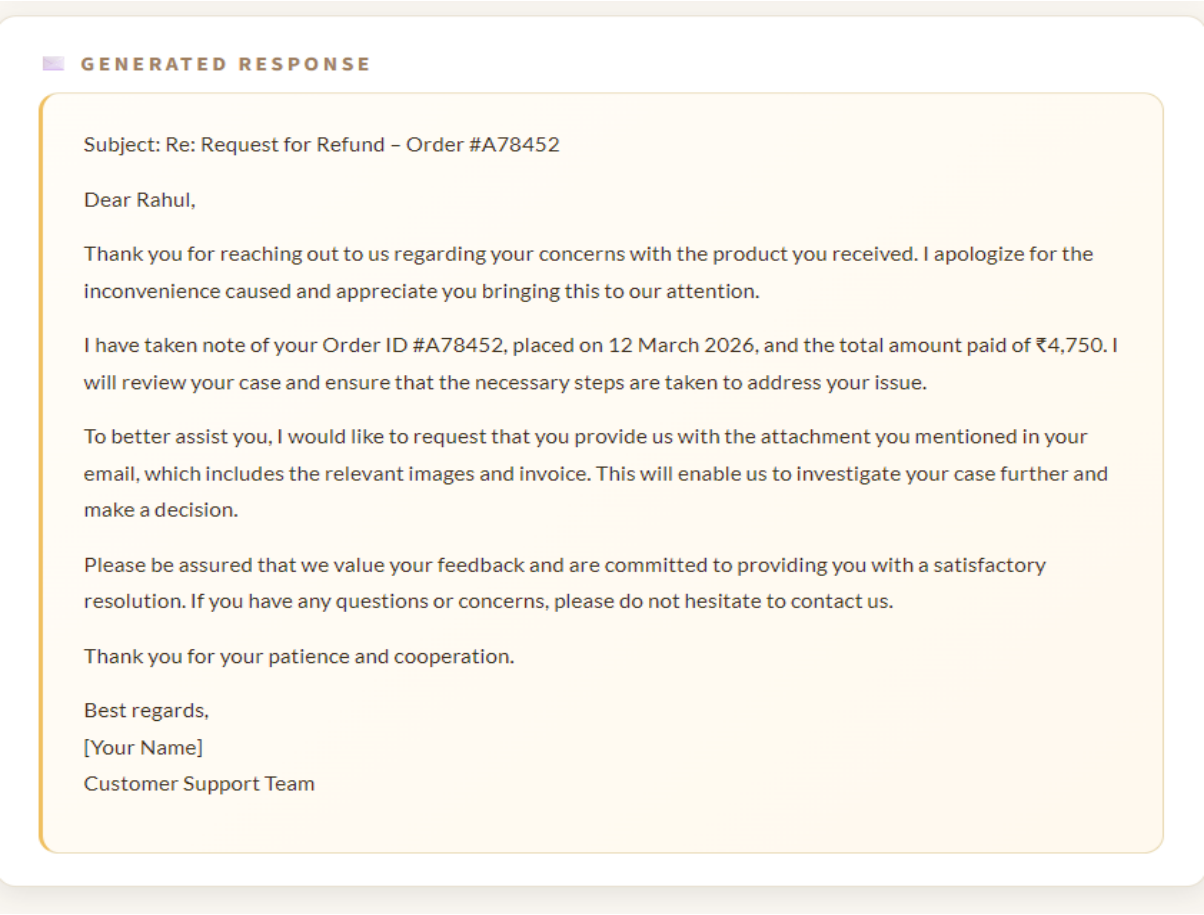


3. Process Request



4. Generate Response





5. Sent to the Sender

T

triageagent5... 11:18

to me

Subject: Re: Request for Refund – Order #A78452

Dear Rahul,

Thank you for reaching out to us regarding the issue with your recent purchase, Order #A78452, which was placed on 12 March 2026. I apologize for the inconvenience caused by the defective product that did not match the description on our website.

I have taken note of your concerns and the attached images and invoice for reference. I will ensure that your request is thoroughly reviewed by our team to determine the next course of action.

You will receive an update on the progress of your request within the next 3-5 working days. If we require any additional information from you, we will be in touch.

Thank you for your patience and cooperation. Please feel free to contact us if you have any further questions or concerns.

Show quoted text

2. Reports / Models

- The system uses an AI-driven triage model to process unstructured email data
- NLP techniques are applied for intent classification and entity extraction
- A generative AI model is used to create context-aware and meaningful responses
- The system operates in real time without dependency on stored datasets

3. Key Outcomes

- Successful automation of email triaging for non-profit organizations
- Significant reduction in manual effort and processing time
- Accurate classification of intent and urgency using AI models
- Faster, consistent, and professional response generation
- End-to-end automation of email communication workflow
- Seamless integration of AI processing with real-world email systems
- Improved handling of high volumes of incoming messages
- Practical demonstration of Generative AI in real-world scenarios

Conclusion

The Non-Profit Triage Support Agent demonstrates the effective use of Artificial Intelligence and Natural Language Processing in automating the handling of unstructured email communication within non-profit organizations. The system addresses a real-world challenge by reducing manual effort, minimizing errors, and improving response efficiency.

By integrating components such as intent classification, urgency detection, entity extraction, and AI-based response generation into a unified pipeline, the system ensures accurate analysis and timely, professional responses. The use of Large Language Models (LLaMA via Groq API) enables context-aware and human-like communication, while the rule-based decision engine enhances reliability and consistency.

The system processes emails in real time and integrates seamlessly with email services, enabling automated retrieval and response delivery. This significantly improves operational efficiency and ensures better prioritization of urgent requests.

Overall, the project highlights the potential of AI-driven solutions in transforming traditional communication workflows into intelligent, automated systems capable of handling high-volume interactions effectively.

Future Scope & Enhancements

- Integration with databases for storing and analyzing email history
- Implementation of multi-language support for better accessibility
- Deployment on cloud platforms for scalability and real-world usage
- Integration with additional communication channels such as SMS and chatbots
- Development of an analytics dashboard for performance monitoring
- Fine-tuning of AI models to improve accuracy and understanding
- Implementation of security and authentication mechanisms
- Automated background processing for continuous email monitoring .